

— INSIGHT · DELIVERY APPROACHES

# **MODULAR VS. TRADITIONAL DATA CENTER DELIVERY: COMPARE THE INTERFACES, NOT THE LABELS**

The useful question is not whether modular delivery is always better than site-built construction. It is which work should happen in a fabrication environment, which work should remain at the site and how the two will be designed, delivered, connected, commissioned and maintained as one system.

## **DEFINE BOTH APPROACHES CLEARLY**

In a modular approach, selected assemblies or systems are fabricated away from the final site, transported, set and connected to site-built infrastructure. In a traditional site-built approach, more of the installation occurs at the project location.

Most real programs are not purely one or the other. They combine off-site and site work in proportions defined by the technical requirements and delivery plan.

## **COMPARE THE DECISIONS THAT**

# CHANGE

| DECISION      | MODULAR EMPHASIS                                                | SITE-BUILT EMPHASIS                                         |
|---------------|-----------------------------------------------------------------|-------------------------------------------------------------|
| Design timing | Module boundaries and interfaces must mature before fabrication | Design can sometimes progress with different package timing |
| Site activity | Site preparation may proceed while fabrication advances         | More installation effort remains at the site                |
| Logistics     | Transport, route, lifting, setting and connection are central   | Material flow, storage and field access are central         |
| Interfaces    | Module-to-site connections need explicit ownership              | Trade and package interfaces remain primarily in the field  |
| Quality plan  | Factory and site verification must connect                      | Field inspection and installation records dominate          |
| Commissioning | Factory work cannot replace final integrated site acceptance    | Testing occurs mainly after site installation               |
| Expansion     | Future module and infrastructure interfaces need definition     | Future field expansion conditions need definition           |

This comparison describes planning considerations—not guaranteed schedule, cost or quality outcomes.

## WHEN MODULAR DESERVES SERIOUS EVALUATION

Modular delivery may be appropriate when the project can define repeatable or bounded assemblies, the design can mature at the required time, transportation and setting are practical, and the site can be prepared to receive the work.

It may be a poor fit when critical requirements remain fluid, the logistics cannot support the assemblies, site interfaces are more complex than the factory scope, or the acceptance plan is not clear.

# DO NOT MOVE THE RISK OUT OF SIGHT

Off-site fabrication changes where work happens. It does not eliminate design coordination, procurement, construction, commissioning or maintenance. Teams should pay particular attention to:

- The freeze point for module design.
- Who owns each factory and site interface.
- How substitutions and changes are controlled.
- What is verified before shipping and after connection.
- How site readiness is measured.
- What documentation follows the module.
- How operating teams access and service the installed system.

## WHERE EVOLVE FITS

Evolve provides self-performed modular fabrication. Modular systems can be factory-built and site-set in days, depending on the project and project plan. Evolve also supports the wider Plan, Design, Build, Power and Maintain lifecycle. Schedule and savings outcomes depend on project conditions and scope.

**EXPLORE MODULAR DATA CENTERS**

## DIRECT ANSWERS

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### IS MODULAR DELIVERY ALWAYS FASTER?

No. It can allow factory and site work to overlap, but design maturity, procurement, transport, site readiness, connections and commissioning still control the program.

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### DOES FACTORY TESTING REPLACE SITE COMMISSIONING?

No. Factory verification can provide useful evidence, but the complete installed system still needs acceptance against the project requirements.

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## WHAT SHOULD AN OWNER DECIDE FIRST?

Define the system boundaries, interfaces, logistics, design-freeze timing, site work and acceptance process before comparing delivery outcomes.

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**Prepared by Evolve Data Center Solutions.** Market sources retrieved September 7, 2026. Market facts are attributed to their sources and are not Evolve performance claims.

### SOURCES

- [JLL, 2026 Global Data Center Outlook](#) (January 2026)